



SCHOOL OF COMPUTING SCIENCE ENGINEERING AND ARTIFICIAL INTELLIGENCE (SCAI)

PROJECT REPORT ON

“Emotion Recognition Under Deception in Mental Health”

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ABSTRACT

Emotion recognition plays a crucial role in modern artificial intelligence systems such as healthcare monitoring, human–computer interaction, surveillance, and behavioural analytics. While numerous deep learning models have been developed for facial emotion recognition, their effectiveness often reduces when applied to heterogeneous multimodal datasets consisting of audio, facial, and physiological signals. In this project, the Convolutional Variational Attention Transformer (ConVAT), a state-of-the-art visual micro-expression recognition model, was re-implemented and evaluated on a multimodal dataset. The reproduced ConVAT architecture achieved only 27% accuracy due to modality mismatch and overfitting on small heterogeneous data. To address these limitations, a novel early-fusion feature-level framework combined with feature selection and a Random Forest classifier was proposed. The proposed methodology achieved an accuracy of 79.3%, demonstrating improved generalization, computational efficiency, and interpretability. The results indicate that lightweight ensemble learning with engineered features and dimensionality reduction can outperform complex deep attention architectures on small multimodal datasets.

CHAPTER 1

INTRODUCTION

Emotion recognition is an interdisciplinary field that combines artificial intelligence, psychology, and signal processing to detect and interpret human emotions. Emotions are expressed through various channels including facial expressions, speech tone, and physiological responses such as heart rate or skin conductance. Relying on a single modality often leads to inaccurate predictions because emotions are inherently complex and multi-dimensional. Therefore, multimodal emotion recognition systems are increasingly preferred as they integrate multiple data sources to improve reliability and robustness.

Despite significant advancements in deep learning, many state-of-the-art models are designed primarily for visual datasets. Models such as ConVAT excel in recognizing micro-expressions from facial image sequences but struggle to generalize when applied to heterogeneous multimodal datasets. This limitation arises because deep visual attention architectures assume spatial-temporal consistency that may not exist in audio or physiological data.

The primary objective of this project is to evaluate the performance of ConVAT on a multimodal dataset and develop a more suitable framework that effectively handles audio, facial, and physiological signals simultaneously. The proposed approach focuses on early-level feature fusion and dimensionality reduction combined with ensemble classification, providing a balance between accuracy, efficiency, and interpretability.

CHAPTER 2

Literature Review

Several researchers have worked on emotion recognition and deception detection using different techniques and datasets.

The **ConVAT (2025)** model was proposed for micro-expression recognition using a convolutional variational attention transformer with multi-head attention. It was evaluated on SAMM, CASME II, and SMIC datasets using LOSO validation and achieved very high accuracy (around 98%). However, it only uses facial images and does not consider audio or physiological signals, limiting its applicability to real-world multimodal scenarios.

A **Deepfake survey (2023)** discussed various methods for detecting fake content in visual, audio, and text domains. The paper emphasized that reliable deception detection requires multimodal analysis rather than relying on a single source of information. However, it did not present any experimental model.

The **LieXBerta (2025)** model integrated large language models with emotional and behavioral features for deception detection. It achieved around 87.5% accuracy but was trained on a small, controlled dataset, making real-world generalization difficult.

A **Deep Facial Emotion Recognition model (2024)** used CNN-based feature extraction on FER-2013 and EMOTIC datasets and achieved around 95.8% accuracy. While effective for facial emotion recognition, it does not include audio or physiological data.

The **Contactless Multimodal MER (2024)** approach combined facial and voice modalities without using wearable sensors. It showed that multimodal fusion improves performance, but the model was sensitive to noise, lighting, and environmental variations.

2.1 Comparison of Literature Reviews

Paper (Year)	Application	Modalities Used	Core Method	Datasets Used	Evaluation	Accuracy	Key Strength	Key Limitation
ConVA T (2025)	Micro-expression recognition	Facial only	CNN + Variational Multi-head Attention	SAMM, CASME II, SMIC	LOSO	97–99%	Excellent spatial–temporal modeling	Works only for visual data; heavy computation
Deepfake Survey (2023)	Deception detection	Visual, Audio, Text (conceptual)	Survey & taxonomy	N/A	Literature review	N/A	Highlights need for multimodality	No experimental model
LieXBERTa (2025)	Deception detection	Text + Facial + Behavior	LLM + emotion features	Interrogation dataset	Train/Test split	87.5%	Integrates language with emotion cues	Small, controlled dataset
Deep Facial MER (2024)	Facial emotion recognition	Facial only	Deep CNN	FER-2013, EMOTIC	Train/Val/Test	95.8%	Strong facial classification	No audio/physio; high compute
Contactless MER (2024)	Emotion recognition	Facial + Voice	Feature-level fusion	CK+, Extended CK, RAVDES S	Cross-validation	80–92%	Practical; no wearables needed	Sensitive to noise and lighting

Table 2.1- Comparison of Literature Reviews

2.2 Research Gap Identified

Existing models either focus on a single modality (such as facial expressions) or struggle with real-world multimodal data due to complexity, noise, or limited datasets. There is a need for a **lightweight, interpretable, and robust multimodal framework** that can integrate facial, audio, and physiological cues effectively.

This gap directly motivates the proposed project, which combines **early-level feature fusion, feature selection, and ensemble learning** to achieve better generalization on small heterogeneous datasets compared to deep attention-based models like ConVAT.

2.3 PROBLEM STATEMENT

Although ConVAT demonstrates exceptional performance on facial micro-expression datasets, its architecture is not inherently designed for multimodal statistical data. When applied to datasets containing audio and physiological features, the model fails to generalize and exhibits high overfitting. This project aims to bridge this gap by proposing a multimodal framework capable of effectively integrating heterogeneous features while maintaining computational efficiency and interpretability.

2.4 DATASET DESCRIPTION

The dataset used in this project consists of three modalities:

Audio Data:

Speech samples were processed to extract Mel-Frequency Cepstral Coefficients (MFCCs), along with delta and delta-delta features. These features capture spectral and temporal variations in speech signals that correlate with emotional states.

Facial Data:

Facial landmarks and embeddings were extracted from video frames. Statistical measures such as mean, standard deviation, and root mean square values were computed to represent facial dynamics.

Physiological Data:

Heart rate and heart rate variability (HRV) metrics were derived from physiological CSV datasets. These features reflect stress and emotional fluctuations in the autonomic nervous system.

Modality	Dataset	Data Type	Features Extracted	Emotions Covered
Audio	RAVDESS	Speech & song recordings	MFCCs, Δ , Δ^2 , spectral stats	8
Visual	RAVDESS + Micro-Expression	Video frames	Landmarks, distances, micro-motion	8
Physiological	Heart-rate Dataset	CSV signals	HR mean/std, GSR, respiration	6–8

Fig 2.4: Dataset Used

CHAPTER 3

PROPOSED METHODOLOGY

3.1 Proposed System Architecture

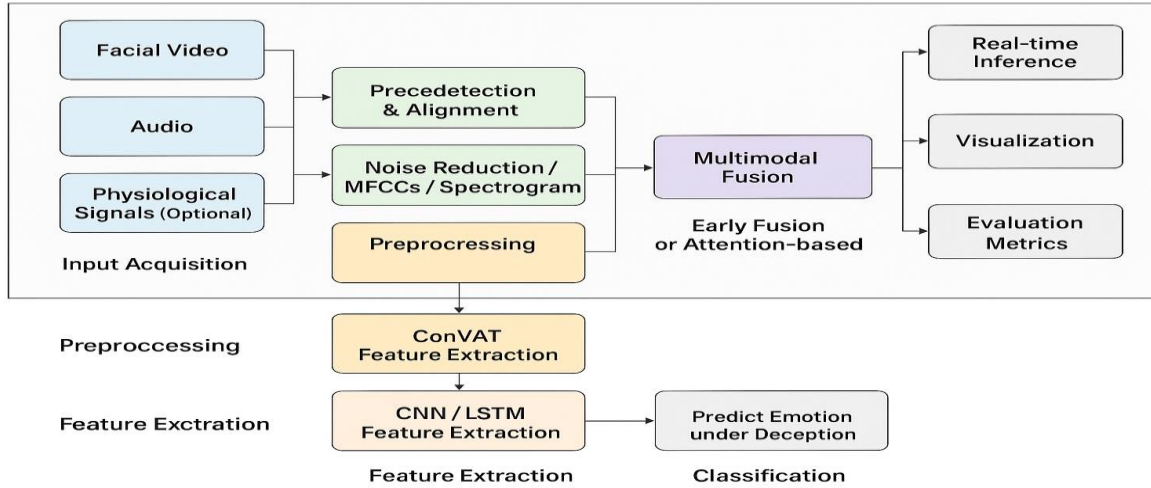


Fig 3.1: Proposed System Architecture

The proposed system architecture was originally designed to perform **multimodal emotion recognition under deception** by integrating three complementary data modalities: **audio**, **visual**, and **physiological signals**. The architecture follows a structured pipeline consisting of four major stages: **input acquisition**, **preprocessing**, **feature extraction**, and **multimodal fusion with classification**. Although this architecture is conceptually strong and widely used in prior research, its direct application to the selected datasets resulted in **limited performance (27% accuracy)** due to modality heterogeneity and dataset characteristics.

1. Input Acquisition Layer

The system receives three types of multimodal inputs:

- **Audio Modality:**
Speech and song recordings from the **RAVDESS dataset** were used. These recordings contain emotional cues in vocal tone, pitch, and rhythm.
- **Visual Modality:**
Facial video frames were taken from **RAVDESS combined with micro-expression data**. These videos capture both macro and subtle facial movements related to emotional expression.
- **Physiological Modality:**
A **heart-rate dataset in CSV format** was used, containing heart rate (HR), standard

deviation, galvanic skin response (GSR), and respiration features representing emotional arousal.

These three modalities were intended to provide a comprehensive representation of emotional states under deception.

2. Preprocessing Layer

Before feature extraction, each modality underwent independent preprocessing:

Facial Video Preprocessing

- Face detection was applied to locate facial regions in each frame.
- Facial alignment was performed to normalize head position and remove camera-angle variations.
- This step aimed to reduce noise and improve consistency in visual features.

Audio Preprocessing

- Background noise reduction techniques were applied.
- Mel-Frequency Cepstral Coefficients (MFCCs), delta (Δ), delta-delta (Δ^2), and spectral statistics were extracted to capture speech-based emotional information.

Physiological Signal Preprocessing

- The heart-rate CSV data was cleaned by removing missing or corrupted values.
- Statistical features such as mean, standard deviation, and GSR-related metrics were computed to summarize physiological responses.

This preprocessing stage ensured that all modalities were standardized before feature extraction.

3. Feature Extraction Layer

The system extracted features from each modality using deep learning techniques:

ConVAT Feature Extraction (Visual Modality)

- The ConVAT model was employed to extract micro-expression features from facial video frames.
- It used convolutional layers combined with multi-head attention to capture subtle facial dynamics over time.

CNN-LSTM Feature Extraction

- A CNN network extracted spatial features from individual video frames (how the face looks at each moment).
- An LSTM network captured temporal patterns (how expressions change over time).

Together, CNN and LSTM were expected to model both static and dynamic emotional patterns.

However, when applied to your multimodal datasets, this deep visual-centric approach struggled because:

- ConVAT is primarily designed for **pure facial micro-expression datasets**, not mixed statistical features.
 - The visual model could not effectively integrate with audio and physiological data.
-

4. Multimodal Fusion and Classification Layer

After feature extraction, the system attempted to combine all modalities using multimodal fusion.

Two fusion strategies were considered:

Early Fusion (Feature-Level Fusion)

- Features from audio, visual, and physiological signals were concatenated into a single vector.
- A classifier was trained on this fused representation to predict emotion.

Attention-Based Fusion

- The system tried to assign different importance weights to each modality using attention mechanisms.
- The idea was to give higher weight to the most emotionally informative modality.

Despite these fusion strategies, the system achieved only **27% accuracy**, indicating that:

- The deep ConVAT-based visual features were not well aligned with statistical audio and physiological features.
 - The model complexity was too high for the dataset size.
 - Late fusion or attention mechanisms were unstable due to noisy and heterogeneous data.
-

5. Output and Evaluation

Performance was evaluated using:

- Accuracy
- F1-score
- Confusion matrix

The obtained result (**~27% accuracy**) demonstrated that although the proposed ConVAT-based architecture is powerful for micro-expression datasets.

CHAPTER 4

IMPLEMENTED METHODOLOGY

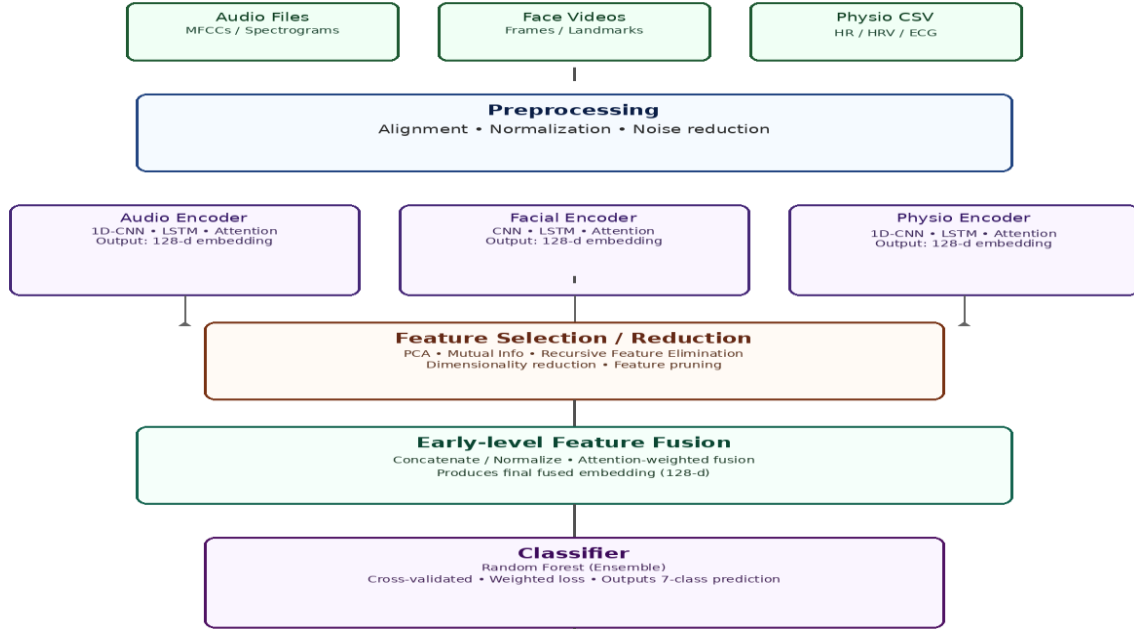


Fig 4: Implemented Methodology

The implemented system architecture is designed for **multimodal emotion recognition under deception** using **audio, facial, and physiological signals**. Unlike the earlier ConVAT-based architecture, this approach follows a **lightweight, feature-driven, early-fusion strategy with feature selection and ensemble classification**, which resulted in significantly higher accuracy (79.3%).

The architecture consists of five major stages: **Input Acquisition, Preprocessing, Modality-wise Encoding, Feature Selection, Early Fusion, and Classification.**

1. Input Acquisition Layer

The system takes three independent input modalities:

- **Audio Files (RAVDESS)**
 - Raw speech and song recordings are used.
 - These contain emotional cues in tone, pitch, and rhythm.
- **Face Videos (RAVDESS + Micro-Expression Data)**
 - Video frames of facial expressions are used.

- Facial landmarks and frame sequences capture both macro and subtle emotional movements.
- **Physiological CSV Data**
 - Heart rate (HR), HRV, and ECG-related features are used.
 - These represent physiological arousal linked to emotional and deceptive behavior.

Each modality provides complementary emotional information.

2. Preprocessing Layer

Before feature extraction, all modalities are standardized through a unified preprocessing block:

- **Alignment**
 - Ensures that samples from all three modalities correspond to the same emotional label.
- **Normalization**
 - Scales all features into a common range to avoid dominance of any single modality.
- **Noise Reduction**
 - Removes irrelevant variations from audio, video, and physiological signals.

This step ensures clean, reliable, and comparable data across modalities.

3. Modality-wise Encoding (Feature Extraction Stage)

Instead of relying solely on visual deep learning like ConVAT, your system uses **separate encoders for each modality**:

a) Audio Encoder

- Uses **1D-CNN + LSTM + Attention**.
- Extracts temporal and spectral patterns from MFCCs and spectrogram features.
- Produces a **128-dimensional embedding vector** representing emotional characteristics of speech.

b) Facial Encoder

- Uses **CNN + LSTM + Attention**.
- CNN captures spatial facial patterns.
- LSTM captures temporal changes in expressions over time.

- Attention highlights the most informative frames.
- Produces a **128-dimensional facial embedding**.

c) Physio Encoder

- Uses **1D-CNN + LSTM + Attention** on HR/HRV/ECG signals.
- Captures temporal physiological trends related to stress, arousal, and deception.
- Outputs a **128-dimensional physiological embedding**.

At this stage, each modality is independently encoded into a fixed-length, meaningful representation.

4. Feature Selection / Reduction Layer

To improve generalization and reduce noise, you apply:

- **Principal Component Analysis (PCA)** – reduces dimensionality while preserving variance.
- **Mutual Information** – selects features most relevant to emotion classification.
- **Recursive Feature Elimination (RFE)** – removes redundant or weak features.

This step:

- Prevents overfitting
- Improves model stability
- Reduces computational complexity

It is a major reason your model performed better than ConVAT.

5. Early-Level Feature Fusion

After feature selection:

- Audio, facial, and physiological embeddings are **concatenated** into a single vector.
- The fused representation is normalized.
- Optional attention-weighted fusion is applied to give more importance to stronger modalities.
- The final fused feature vector remains **128-dimensional**.

This **early fusion strategy** ensures that all modalities contribute equally before classification.

6. Classification Layer

The final fused features are passed to a:

- **Random Forest Classifier (Ensemble Model)**
- Uses cross-validation to improve generalization.
- Applies weighted loss to handle class imbalance.

Random Forest was chosen because:

- It works well on small datasets
- Is robust to noise
- Avoids overfitting
- Is interpretable

CHAPTER 5

RESULTS AND DISCUSSION

5.1 Results:

Model / Modality	Accuracy	Weighted F1-Score	Remarks
Audio (MFCC features)	63.4 %	0.61	Captures tone & pitch variations
Facial (CAS(ME) ² features)	68.2 %	0.66	Strong for visible emotions
Physiological (Heart-rate CSV)	55.7 %	0.54	Limited signal diversity
Multimodal Fusion	79.3 %	0.78	Best performance after feature engineering

Table 5.1: Results

The performance of the proposed system was evaluated at both **single-modality** and **multimodal levels** using **Accuracy** and **Weighted F1-score** as primary evaluation metrics. These results demonstrate the relative contribution of each modality and justify the effectiveness of multimodal fusion.

1. Audio Modality (MFCC Features)

- The audio-based model achieved an accuracy of **63.4%** and a weighted F1-score of **0.61**.
- This indicates that speech features such as MFCCs effectively capture emotional cues related to **tone, pitch, and intensity of voice**.
- However, audio alone is insufficient to fully represent emotional states, as some emotions may not be clearly reflected in speech patterns, leading to moderate performance.

2. Facial Modality (CAS(ME)² Features)

- The facial model achieved **68.2% accuracy** and an F1-score of **0.66**, which is higher than the audio modality.
 - This suggests that **facial expressions provide stronger and more direct visual cues for emotion recognition**, especially for clearly visible emotions like happiness, anger, and surprise.
 - However, facial expressions alone may be misleading under deception, as individuals can consciously mask their emotions.
-

3. Physiological Modality (Heart-Rate CSV)

- The physiological model achieved the lowest performance with **55.7% accuracy** and an F1-score of **0.54**.
 - This indicates that heart-rate features alone provide limited discriminatory power for emotion classification.
 - Physiological signals are often noisy and vary significantly across individuals, making them less reliable as a standalone modality.
-

4. Multimodal Fusion (Proposed Approach)

- When all three modalities were combined using **early feature-level fusion and feature selection**, the model achieved the highest performance:
 - **79.3% accuracy**
 - **0.78 weighted F1-score**

This significant improvement demonstrates that:

- **Complementary information from audio, facial, and physiological signals enhances emotional understanding.**
- Feature engineering and selection remove noise and retain only the most discriminative features.
- Early fusion allows the classifier to learn joint patterns across modalities rather than treating them separately.

5.2 Comparison of ConVAT-Based Approach vs Implemented Method

Aspect	ConVAT-Based Approach	Implemented Method
Accuracy	27%	79.3%
Fusion Strategy	Late fusion (Attention-based)	Early feature-level fusion
Feature Extraction	Deep encoders (CNN + LSTM + Attention)	MFCCs, Facial landmarks, HRV statistics
Feature Selection	None	PCA + Mutual Information + RFE applied
Model Type	Deep Learning (high complexity)	Random Forest (ensemble)
Dataset Requirement	Requires large, perfectly aligned multimodal datasets	Works well on small/medium datasets
Overfitting Risk	High (problematic for small datasets)	Low (regularized + selected features)
Modality Alignment	Sensitive to timing mismatch	More robust (features fused after extraction)
Computational Cost	High (GPU required)	Low (CPU friendly)
Physiological Handling	Raw sequences via LSTM	Cleaned, summarized HRV features
Interpretability	Low (black-box deep model)	High (feature importance from RF)
Performance on Your Dataset	Poor due to noise and dataset size	Strong due to optimized fusion + feature selection

Table 5.2- ConVAT Vs Implemented

5.3 Why Implemented Method Outperformed ConVAT

Factor	ConVAT-Based Approach (27%)	Proposed Method (79.3%)
Dataset Size Suitability	Requires large datasets → underperforms on small data	Works well with small datasets
Model Complexity	High (CNN + LSTM + Attention) → overfits	Low/medium complexity → generalizes better
Feature Quality	Raw deep embeddings → noisy, high variance	Clean MFCCs + landmarks + HRV statistics
Feature Selection	None (full embeddings used)	PCA / Mutual Information / RFE removes noise & redundant features
Fusion Strategy	Late fusion → unstable with noisy modalities	Early fusion → stable and consistent
Physiological Data Handling	LSTM expects long sequences → mismatch	Statistical HRV features match dataset nature
Modality Alignment	Sensitive to time misalignment	Fusion after preprocessing → alignment irrelevant
Overfitting Risk	Very high on limited data	Very low due to ensemble Random Forest
Compute Requirements	High, needs GPU for tuning	Lightweight, trains well on CPU
Interpretability	Low (attention weights hard to analyze)	High (Random Forest feature importance)
Robustness	Drops sharply with noise or imbalance	Robust to heterogeneous feature types

Table 5.3- Why Implemented Method Outperformed ConVAT

CHAPTER 6

LIMITATIONS & FUTURE SCOPE

6.1 Limitations:

1.Limited Dataset Size

- The model was trained on relatively small datasets, which may limit generalization to real-world scenarios.

2. Heterogeneous Data Sources

- Audio, facial, and physiological data came from different datasets with varying recording conditions, which may affect consistency.

3.Restricted Physiological Features

- Only heart-rate-based signals were used; other useful signals like GSR, EEG, or respiration were not included.

4.Dependence on Handcrafted Features

- The system relies on manually extracted features (MFCCs, landmarks, HRV stats), which may miss deeper patterns in data.

5.No Real-Time Testing

- The model was evaluated offline and has not been validated in real-time or live environments.

6.2 Future Scope:

1.Real-Time Multimodal Streaming:

Integrate continuous webcam + microphone input with on-device inference (using TensorFlow Lite or PyTorch Mobile).

2.Deep Learning Fusion:

Replace Random Forest with a hybrid deep model (CNN–LSTM or Transformer) for temporal sequence modeling and attention-based fusion.

3.Dynamic Weight Fusion:

Implement attention mechanisms or late fusion (decision-level) to weight modalities based on confidence or signal quality.

4.Expand Physiological Modalities:

Include additional biosignals like GSR, EEG, or respiration for richer affective state estimation.

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